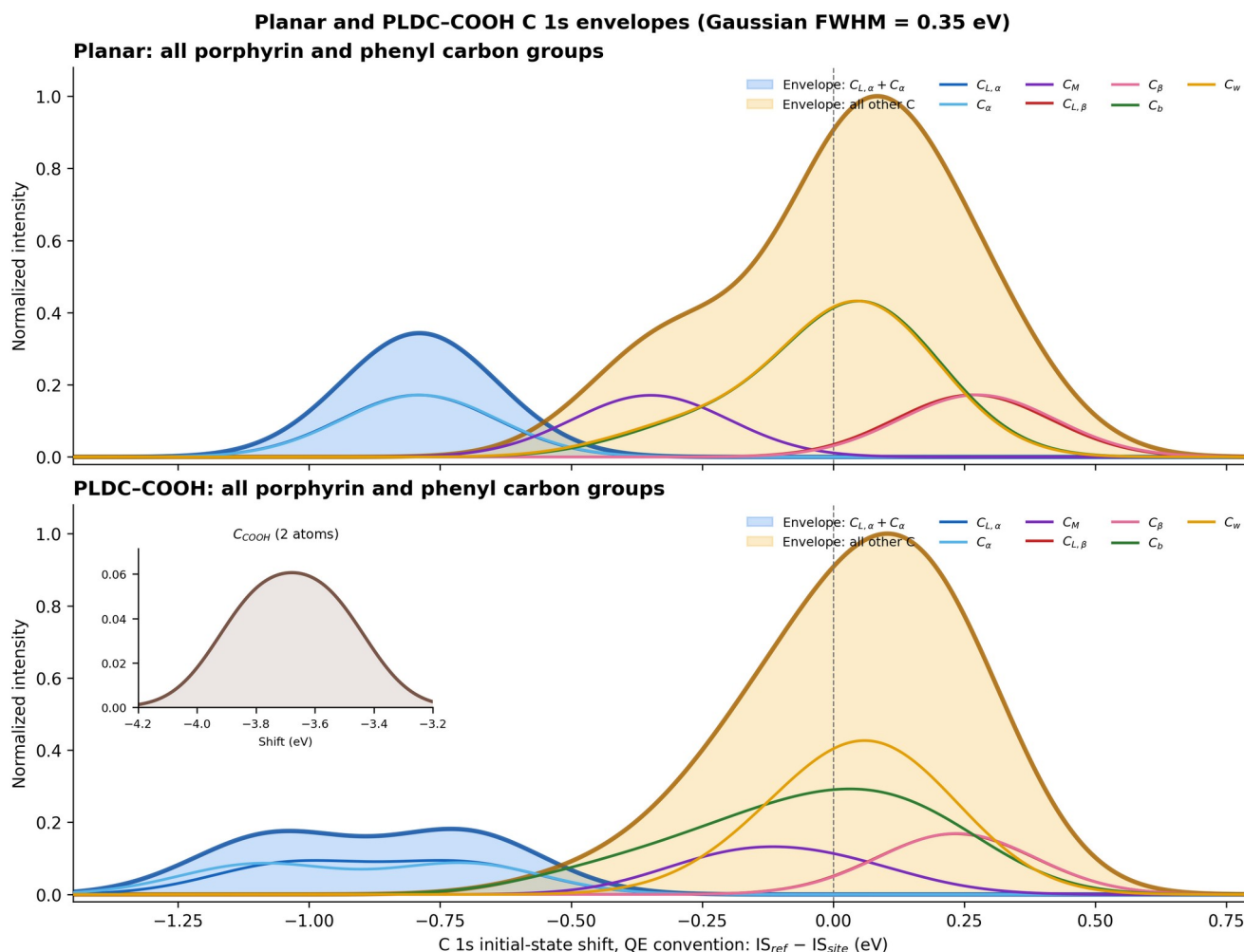


# Planar versus Protonated PLDC

## All-Carbon Core-Level Shifts

Initial-state Quantum ESPRESSO comparison · all carbon groups

1 August 2026



### Result summary

The all-carbon Planar dataset and a new neutral protonated PLDC calculation are complete. The PLDC model contains two COOH groups, with H82 bonded to O78 and H83 bonded to O80. All 44 Planar and all 46 PLDC carbon atoms were evaluated.

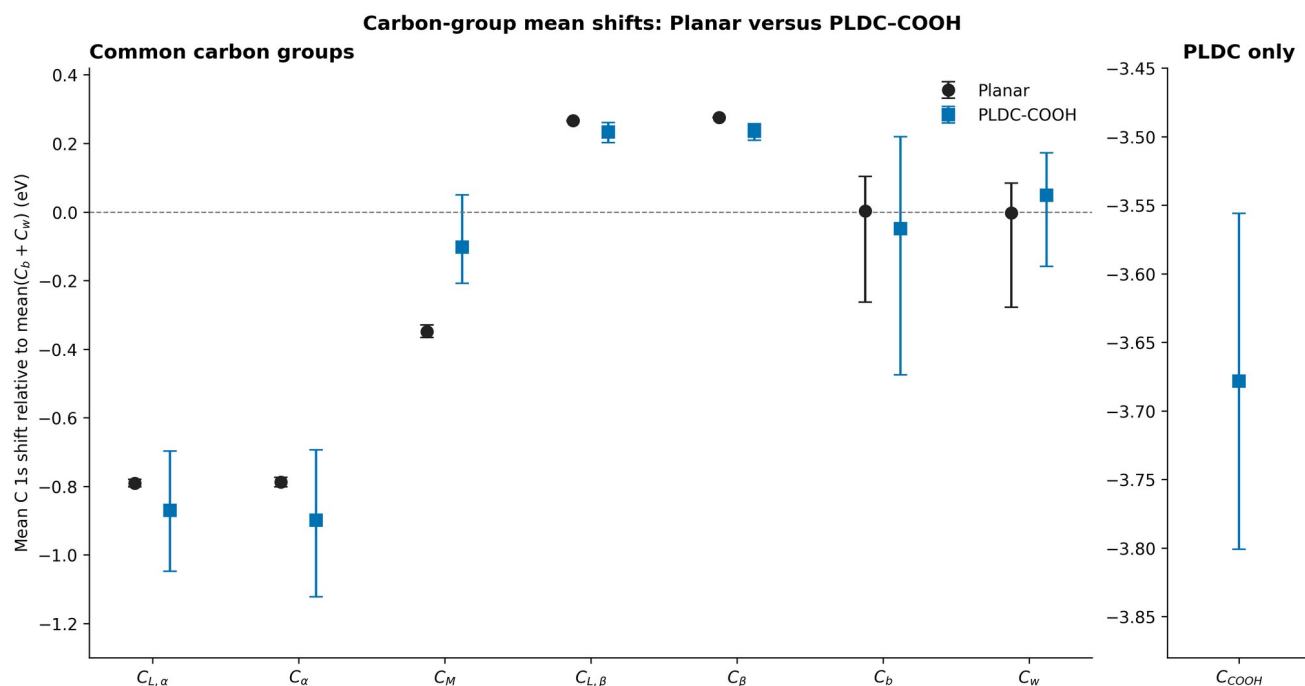
Every atom contributes a unit-area Gaussian with FWHM = 0.35 eV. The blue background is the combined  $C_{L,\alpha}$  +  $C_{\alpha}$  envelope; the contrasting amber background contains all remaining carbon groups. Individual group curves are drawn in the foreground, and no total curve is shown. The CSV files contain the discrete Quantum ESPRESSO values.

## Carbon-group shifts

Shifts use the Quantum ESPRESSO example convention:  $\Delta E = \text{IS}(\text{reference}) - \text{IS}(\text{site})$ . The internal reference is the mean initial-state contribution of all Cb and Cw atoms in the same molecule.

| Group     | Planar n | Planar mean | PLDC n | PLDC mean | PLDC range (eV)  |
|-----------|----------|-------------|--------|-----------|------------------|
| C_L,alpha | 4        | -0.791      | 4      | -0.870    | -1.048 to -0.697 |
| C_alpha   | 4        | -0.788      | 4      | -0.899    | -1.122 to -0.694 |
| C_M       | 4        | -0.348      | 4      | -0.102    | -0.208 to 0.050  |
| C_L,beta  | 4        | 0.266       | 4      | 0.233     | 0.202 to 0.261   |
| C_beta    | 4        | 0.275       | 4      | 0.235     | 0.210 to 0.257   |
| C_b       | 12       | 0.003       | 12     | -0.048    | -0.475 to 0.220  |
| C_w       | 12       | -0.003      | 12     | 0.048     | -0.158 to 0.172  |
| C_COOH    | —        | —           | 2      | -3.678    | -3.801 to -3.556 |

The two PLDC C\_COOH atoms form a separated feature with mean shift  $-3.678$  eV (range  $-3.801$  to  $-3.556$  eV). This sign follows the Quantum ESPRESSO initial-state convention above.



## Calculation details

|                             |                                                                                                                                                          |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Structures</b>           | Planar ZnC <sub>44</sub> H <sub>28</sub> N <sub>4</sub> (77 atoms); PLDC–COOH ZnC <sub>46</sub> H <sub>28</sub> N <sub>4</sub> O <sub>4</sub> (83 atoms) |
| <b>Electronic model</b>     | Neutral, closed-shell single-point calculations                                                                                                          |
| <b>Quantum ESPRESSO</b>     | Version 7.5; pw.x followed by initial_state.x                                                                                                            |
| <b>Exchange–correlation</b> | PBE                                                                                                                                                      |
| <b>Cell / isolation</b>     | 25 Å cubic cell; Martyna–Tuckerman isolated correction; $\Gamma$ point                                                                                   |
| <b>Plane-wave cutoffs</b>   | 30 Ry wave functions; 180 Ry charge density                                                                                                              |
| <b>Occupations</b>          | Marzari–Vanderbilt smearing, 0.02 Ry                                                                                                                     |
| <b>Convergence</b>          | conv_thr = $1 \times 10^{-6}$ Ry; local-TF mixing, $\beta = 0.20$                                                                                        |
| <b>Execution</b>            | 4 MPI processes                                                                                                                                          |

## Convergence and runtime

|                                  |                                                                         |
|----------------------------------|-------------------------------------------------------------------------|
| <b>Planar SCF</b>                | 32 iterations; −739.91431933 Ry; 15 min 05 s wall                       |
| <b>Planar initial_state.x</b>    | 13.14 s wall                                                            |
| <b>PLDC–COOH SCF</b>             | 40 iterations; −890.04265055 Ry; final accuracy $5.6 \times 10^{-7}$ Ry |
| <b>PLDC–COOH SCF runtime</b>     | 1 h 38 min wall                                                         |
| <b>PLDC–COOH initial_state.x</b> | 1 min 06 s wall                                                         |

## Important modeling note

The two O–H positions were constructed with 0.98 Å bonds by extending the longer C–O direction in each supplied carboxyl group. No geometry optimization was performed after proton placement. The reported PLDC shifts are therefore single-point values for this constructed neutral geometry; geometry optimization and cutoff/cell convergence should precede publication-quality interpretation.

## Files

|                      |                                                                |
|----------------------|----------------------------------------------------------------|
| <b>inputs/</b>       | PLDC SCF and initial_state.x input files                       |
| <b>outputs/</b>      | Converged pw.x and initial_state.x output files                |
| <b>restart_data/</b> | Quantum ESPRESSO save directory for restarting/post-processing |
| <b>scripts/</b>      | Preparation and all-carbon comparison scripts                  |
| <b>CSV files</b>     | Per-atom Planar, PLDC, combined, and group-summary data        |
| <b>PNG/PDF files</b> | Full-range, zoomed, and group-mean comparison figures          |